5.2 Decoder and Preprocessing

The decoder reconstructs complete length-prefixed frames, extracts available aircraft fields and maintains an ICAO-keyed state snapshot. The preprocessor converts records to a stable schema: latitude/longitude in decimal degrees, altitude in millimetres, heading in degrees multiplied by 100, and velocities in centimetres per second.

5.3 Storage and Export

SQLite stores one latest record per flight and a filtered trajectory history. A new trajectory point is saved when time or configured motion thresholds are exceeded. The storage module also produces daily CSV files, trajectory exports and ICAO-wise workbooks. SQLite is embedded [5], and WAL mode is enabled for the local workload [6].

5.4 User Interface

Leaflet renders map markers, live paths and dotted prediction paths [7]. Chart.js renders altitude history for the selected aircraft [8]. The interface polls aircraft data every second and status data every four seconds, while search, selection, route information and live/simulation controls remain on one dashboard.

5.5 Error Handling and Data Quality

The data source retains incomplete TCP bytes until a complete frame is available. The decoder checks supported layouts and coordinate ranges, the preprocessor handles missing values, and the storage layer rejects zero-position samples and suppresses excessive duplicates. Predicted rows are clearly labelled and are never written back as real observations.

5.6 Configuration Parameters

Parameter	Default / Purpose
simulation_mode	true; chooses captured replay or live receiver mode.
replay_batch	20 frames per browser refresh cycle.
simulation stale_seconds	12 s before replay aircraft leave the active snapshot.
live stale_seconds	45 s before live aircraft leave the active snapshot.
FLIGHT_DB_MIN_INTERVAL_SEC	5 s storage interval threshold.
FLIGHT_PREDICTION_MAX_AGE_SEC	180 s maximum short-outage prediction age.
FLIGHT_DB_HISTORY_RETENTION_DAYS	30 days of local trajectory history.